

Supplementary Information for AI-Driven Structure Refinement of X-ray Diffraction

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Code & Main**GitHub:** <https://github.com/Bin-Cao/PyWPEM>**Manuscript (Main):** <https://arxiv.org/abs/2602.16372v1>[Back to Contents](#) | 3

1 Statistical modeling

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1.1 M-component Mixture Model

Throughout this paper, the following notation is adopted. $Y^{\text{ex}}(x_j)$ denotes the experimentally observed XRD profile, $Y^{\text{nb}}(x_j)$ the measured profile after background removal, and $Y^{\text{WPEM}}(x_j)$ the calculated pattern obtained from an m -component mixture model. Here, x_j ($j = 1, 2, \dots, |\widetilde{2\theta}|$) represents the observation locations corresponding to the diffraction angle $\widetilde{2\theta}$, and $|\widetilde{2\theta}|$ is the total number of sampled points. The ultimate objective of the model is to minimize the residual between $Y^{\text{WPEM}}(x_j)$ and $Y^{\text{ex}}(x_j)$.

The profile fitting task is formulated as a density estimation problem based on an m -component mixture model and is solved using a batch optimization strategy termed the Bragg-based Expectation Maximization process for whole-pattern decomposition. The background-free powder XRD profile is interpreted as a measure function composed of m peak shape functions (PSFs), each representing an underlying diffraction regime. A soft assignment associates each observation $\widetilde{2\theta}$ with all components, yielding

$$Y^{\text{WPEM}}(x_j) = \sum_{i=1}^m W_i Y_i(x_j), \quad (1)$$

where $Y_i(x_j)$ denotes the i -th PSF and W_i is its corresponding weight, reflecting the effective number of observations explained by that component. The location parameter of each PSF determines the diffraction index of the associated Bragg peak and converges to the value prescribed by Bragg's law. Because PSFs are defined as continuous functions, smooth optimization schemes can be employed, enabling a continuous and physically consistent representation of diffraction intensities over the entire angular range.

In our framework, an XRD pattern is constructed as a measure function under the assumption that each component in the mixture model follows a specific probability density function (p.d.f.). Under this formulation, profile fitting is converted into a density estimation problem with latent variables, which is solved using the EM algorithm to perform maximum likelihood estimation subject to the constraint imposed by Bragg's law. The generative process can be described explicitly as follows. A latent variable W is drawn from a discrete p.d.f. w , such that $P(W = W_i) = w_i$, and is assigned to component Y_i . Each component Y_i follows a p.d.f. y_i parameterized by θ , that is, $Y_i \sim y_i(\theta)$. Accordingly, the resulting m -component mixture model can be written as

$$y^{\text{WPEM}}(x_j) = \sum_{i=1}^m w_i y_i(x_j). \quad (2)$$

From this perspective, the XRD pattern is a continuous measure function defined on the field of real numbers, satisfying

$$\Omega(y^{\text{WPEM}}) = \int_{-\infty}^{+\infty} y^{\text{WPEM}}(x) dx = I_{\text{area}}, \quad (3)$$

where I_{area} denotes the total integrated intensity, corresponding to the area enclosed by the background-free XRD profile. Consequently,

$$\int_{-\infty}^{+\infty} \left(\sum_{i=1}^m w_i y_i(x) \right) dx = \Omega(y^{\text{WPEM}}), \quad (4)$$

which yields the normalization condition $\sum_{i=1}^m w_i = I_{\text{area}}$, with $w_i \in (0, I_{\text{area}})$. For XRD patterns defined over the 2θ domain, thin-tailed p.d.f.s are recommended to model the PSF, ensuring both physical plausibility and numerical stability.

Each PSF, described by an individual p.d.f., y_i , is further modeled using a Pseudo-Voigt (PV) function p_{pv} , which itself is a weighted mixture of Lorentzian and Gaussian components. The Lorentzian component captures the intrinsic diffraction emission and the time-of-reflection probability associated with the interaction between a crystal plane and the incident X-ray beam, whereas the Gaussian component accounts for stochastic broadening arising from instrumental effects and random noise. Each PV function is a thin-tailed, continuous p.d.f. satisfies

$$\int_{-\infty}^{+\infty} p_{\text{pv}}(x) dx = 1. \quad (5)$$

It is defined as

$$p_{\text{pv}}(x) = \Delta \cdot \frac{1}{\pi} \frac{\gamma}{(x - \mu)^2 + \gamma^2} + (1 - \Delta) \cdot \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right), \quad (6)$$

where $\Delta \in (0, 1)$ adjusts the angle-dependent noise level and represents the weight of the Lorentzian component. Substituting this p.d.f. into Eq. 2 yields

$$y^{\text{WPEM}}(x) = \sum_{i=1}^m w_i \left[\frac{\Delta_i}{\pi} \frac{\gamma_i}{(x - \mu_i)^2 + \gamma_i^2} + \frac{1 - \Delta_i}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{(x - \mu_i)^2}{2\sigma_i^2}\right) \right], \quad (7)$$

where μ_i denotes the Bragg peak position of the i -th diffraction peak, γ_i is the Half Width at Half Maximum (HWHM) of the Lorentzian distribution, and σ_i^2 is the variance of the Gaussian distribution.

In the Le Bail method and related whole powder pattern decomposition approaches, the number of refinement parameters is reduced by enforcing an identical FWHM (Γ) for the Lorentzian and Gaussian components of the PV function. Specifically,

$$2\gamma = 2\sqrt{2 \ln 2} \sigma = \Gamma, \quad (8)$$

where Γ is refined using a semi-empirical model, for example,

$$\Gamma^2 = U \tan^2 \tilde{\theta} + V \tan \tilde{\theta} + W. \quad (9)$$

In our workflow, the Lorentzian and Gaussian width parameters, γ and σ^2 , are treated as independent variables and are directly optimized within the EM–Bragg process.

1.2 Construction of the Q Function in the EM–Bragg Process

The EM–Bragg process is formulated to maximize the posterior likelihood of the latent variable T_{ij} , where i indexes the PV components and j denotes the sampling index corresponding to $\widehat{2\theta}$. The model parameters are defined as $\{\Delta_i, w_i, \mu_i, \gamma_i, \sigma_i^2\} \triangleq \boldsymbol{\theta}$, the observations as $(x_1, x_2, \dots, x_j) \triangleq \mathbf{X}$, and the corresponding measure function as $y^{\text{WPEM}}(x) \triangleq P$. Under this notation, parameter estimation amounts to maximizing the log-likelihood

$$\arg \max_{\boldsymbol{\theta}} \log P(\mathbf{X} | \boldsymbol{\theta}). \quad (10)$$

To facilitate the optimization, the latent variable T_{ij} is introduced, yielding

$$\arg \max_{\boldsymbol{\theta}} \log \left(\int P(\mathbf{X}, T_{ij} | \boldsymbol{\theta}) dT_{ij} \right), \quad (11)$$

since $P(\mathbf{X} | \boldsymbol{\theta})$ is the marginal distribution of the joint probability $P(\mathbf{X}, T_{ij} | \boldsymbol{\theta})$. Assuming that the distribution of T_{ij} is known as $P(T_{ij})$, Jensen's inequality leads to

$$\begin{aligned} \arg \max_{\boldsymbol{\theta}} \log \left(\int P(\mathbf{X}, T_{ij} | \boldsymbol{\theta}) dT_{ij} \right) &= \arg \max_{\boldsymbol{\theta}} \log \left(\int \frac{P(\mathbf{X}, T_{ij} | \boldsymbol{\theta})}{P(T_{ij})} P(T_{ij}) dT_{ij} \right) \\ &\geq \arg \max_{\boldsymbol{\theta}} \int \log \left(\frac{P(\mathbf{X}, T_{ij} | \boldsymbol{\theta})}{P(T_{ij})} \right) P(T_{ij}) dT_{ij}. \end{aligned} \quad (12)$$

Eq. 12 corresponds to evaluating the expectation of $\log(P(\mathbf{X}, T_{ij} | \boldsymbol{\theta})/P(T_{ij}))$ with respect to the latent variable T_{ij} , which constitutes the E-step of the EM–Bragg process. Here, $P(T_{ij})$ depends only on the parameters obtained at the t -th iteration and is therefore written as $P(T_{ij} | \boldsymbol{\theta}^{(t)})$. Substituting this relation into Eq. 12 yields

$$\begin{aligned} \arg \max_{\boldsymbol{\theta}} \int \log \left(\frac{P(\mathbf{X}, T_{ij} | \boldsymbol{\theta})}{P(T_{ij} | \boldsymbol{\theta}^{(t)})} \right) P(T_{ij} | \boldsymbol{\theta}^{(t)}) dT_{ij} &= \arg \max_{\boldsymbol{\theta}} \left\{ \int \log (P(\mathbf{X}, T_{ij} | \boldsymbol{\theta})) P(T_{ij} | \boldsymbol{\theta}^{(t)}) dT_{ij} \right. \\ &\quad \left. - \int \log (P(T_{ij} | \boldsymbol{\theta}^{(t)})) P(T_{ij} | \boldsymbol{\theta}^{(t)}) dT_{ij} \right\}. \end{aligned} \quad (13)$$

The second term in Eq. 13 is independent of $\boldsymbol{\theta}$ for a fixed $\boldsymbol{\theta}^{(t)}$ and can therefore be omitted during maximization. The remaining term defines the *Q function* in the EM–Bragg process,

$$Q(\boldsymbol{\theta} | \boldsymbol{\theta}^{(t)}) = \int \log (P(\mathbf{X}, T_{ij} | \boldsymbol{\theta})) P(T_{ij} | \boldsymbol{\theta}^{(t)}) dT_{ij}, \quad (14)$$

which constitutes a lower bound on the log-likelihood. Accordingly, the M-step updates the parameters as

$$\boldsymbol{\theta}^{(t+1)} = \arg \max_{\boldsymbol{\theta}} Q(\boldsymbol{\theta} | \boldsymbol{\theta}^{(t)}). \quad (15)$$

At each iteration, the Bragg likelihood is reconstructed using the updated peak positions $\mu_i^{(t+1)}$, and the corresponding parameters are re-optimized, as detailed below.

1.3 Initialization of Model

According to Bragg's law, diffraction occurs when the scattering vector coincides with a reciprocal lattice vector, which can be expressed as

$$2d_{(HKL)} \sin \tilde{\theta}_{\text{Brg}} = \lambda, \quad (16)$$

where $d_{(HKL)}$ denotes the interplanar spacing and (H, K, L) are the Miller indices of the diffracting planes.

The preliminary peak positions $\mu_i^{(0)}$ are computed directly from Bragg's equation using lattice constants obtained from an initial structure identification model, such as Xquyer¹. Based on $\mu_i^{(0)}$, the initial values of the Lorentzian and Gaussian width parameters, $\gamma_i^{(0)}$ and $\sigma_i^{2(0)}$, are estimated from the FWHM, Γ_i , of the diffraction peaks,

$$\gamma_i^{(0)} = \Delta_i^{(0)} \frac{\Gamma_i}{2}, \quad \sigma_i^{2(0)} = \left(1 - \Delta_i^{(0)}\right) \frac{\Gamma_i}{2}, \quad (17)$$

where $\Delta_i^{(0)}$ denotes the Lorentzian mixing weight in the pseudo-Voigt (PV) function. Unless otherwise specified, $\Delta_i^{(0)}$ is set to 0.8, reflecting that the Lorentzian component dominates the PSFs.

To ensure numerical stability during the iterative optimization, both γ_i and σ_i^2 are constrained to remain larger than a small positive constant ε (default $\varepsilon = 5 \times 10^{-4}$), although the precise threshold may be adjusted depending on the application.

The initial peak weights $w_i^{(0)}$ are defined as

$$w_i^{(0)} = \frac{y_i^{\text{nb}}}{\sum_{i=1}^m y_i^{\text{nb}}} I_{\text{area}}, \quad i = 1, \dots, m, \quad (18)$$

where y_i^{nb} denotes the background-eliminated intensity and I_{area} is the total integrated intensity of the diffraction profile.

These initial values are used to initialize the EM–Bragg process, after which reliable results are obtained through posterior likelihood maximization.

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1.4 EM–Bragg Process

In the m -component mixture model, each observation contributes to all components y_i , and the current parameter estimates are used to assign responsibilities of the data to individual diffraction peaks. Notably, there are y_j^{nb} samples at the j -th angular position x_j . During the maximization step, these responsibilities are incorporated into a weighted maximum posterior likelihood estimation together with constraints imposed by Bragg's law.

Accordingly, the posterior probability $P(T_{ij})$ is defined to quantify the responsibility of observation x_j for the i -th peak y_i , given by

$$\begin{aligned} P(T_{ij}) &= \frac{w_i p_{\text{pv},i}(x_j)}{\sum_{i=1}^m w_i p_{\text{pv},i}(x_j)} \\ &= \frac{w_i \left[\frac{\Delta_i}{\pi} \frac{\gamma_i}{(x_j - \mu_i)^2 + \gamma_i^2} + \frac{1 - \Delta_i}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{(x_j - \mu_i)^2}{2\sigma_i^2}\right) \right]}{\sum_{i=1}^m w_i \left[\frac{\Delta_i}{\pi} \frac{\gamma_i}{(x_j - \mu_i)^2 + \gamma_i^2} + \frac{1 - \Delta_i}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{(x_j - \mu_i)^2}{2\sigma_i^2}\right) \right]}, \end{aligned} \quad \begin{aligned} j &= 1, \dots, n, \\ i &= 1, \dots, m. \end{aligned} \quad (19)$$

Here, m denotes the number of mixture components and n the number of observed $\widehat{2\theta}$ sampling points. The latent variable T_{ij} represents the fractional contribution of observation j to the i -th pseudo-Voigt component $p_{pv,i}$, commonly referred to as the responsibility of component i for observation j . Under the soft-assignment assumption, the normalization condition $\sum_{i=1}^m T_{ij} = 1$ holds for each j .

The Q function $Q(\theta \mid \theta^{(t)})$ derived above can be explicitly written for the EM–Bragg process as

$$\begin{aligned} Q(\theta \mid \theta^{(t)}) &= \ln \left[\prod_{j=1}^N \left(\sum_{i=1}^m w_i p_{pv,i}(x_j) \right)^{y_j^{\text{nb}}} \right] \\ &= \sum_{j=1}^N y_j^{\text{nb}} \ln \left(\sum_{i=1}^m w_i p_{pv,i}(x_j) \right) \\ &= \sum_{j=1}^N y_j^{\text{nb}} \ln \sum_{i=1}^m w_i \left[\frac{\Delta_i}{\pi} \frac{\gamma_i}{(x_j - \mu_i)^2 + \gamma_i^2} + \frac{1 - \Delta_i}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{(x_j - \mu_i)^2}{2\sigma_i^2}\right) \right]. \end{aligned} \quad (20)$$

The sufficient statistics in this formulation are given by the pairs $(x_j, T_{ij}^{(t)})$. However, direct maximization of Eq. 20 is not tractable in practice. This difficulty arises from the Lorentzian component in each mixture term, whose expectation do not exist, as the Lorentzian profile corresponds to a Cauchy distribution.

To address this issue, we adopt a mathematical construction to derive a surrogate lower bound, which guarantees a monotonic increase of the log-likelihood lower bound at each iteration, i.e., $Q(\theta^{(t+1)}) > Q(\theta^{(t)})$, thereby ensuring monotonic convergence of the EM–Bragg algorithm. The specific update rules are detailed below.

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1.5 Mathematical Construction

Eq. 19 can be decomposed into Lorentzian and Gaussian contributions, yielding

$$T_{ji}^L = \frac{w_i \frac{\Delta_i}{\pi} \frac{\gamma_i}{(x_j - \mu_i)^2 + \gamma_i^2}}{\sum_{i=1}^m w_i p_{pv,i}(x_j)}, \quad (21)$$

$$T_{ji}^R = \frac{w_i \frac{1 - \Delta_i}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{(x_j - \mu_i)^2}{2\sigma_i^2}\right)}{\sum_{i=1}^m w_i p_{pv,i}(x_j)}. \quad (22)$$

We further define the auxiliary quantity

$$\beta_{ji}^L \triangleq T_{ji}^L \frac{\gamma_i}{\pi [(x_j - \mu_i)^2 + \gamma_i^2]}, \quad j = 1, \dots, n, \quad i = 1, \dots, m. \quad (23)$$

In the maximization step, all parameters are updated by maximizing the Q function given the latent variables $\mathbb{P}(T_{ij}^{(t)})$. This leads to the following constrained optimization problem:

$$\hat{\theta}^{(t+1)} = \arg \max_{\theta} \left\{ \sum_{j=1}^N y_j^{\text{nb}} \ln \sum_{i=1}^m w_i \left[\frac{\Delta_i}{\pi} \frac{\gamma_i}{(x_j - \mu_i)^2 + \gamma_i^2} + \frac{1 - \Delta_i}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{(x_j - \mu_i)^2}{2\sigma_i^2}\right) \right] \right\}, \quad (24)$$

subject to the normalization constraint

$$\sum_{i=1}^m w_i = I_{\text{area}}. \quad (25)$$

Here, $\hat{\theta} = \{\hat{\Delta}_i, \hat{w}_i, \hat{\mu}_i, \hat{\gamma}_i, \hat{\sigma}_i^2\}$, $\theta = \{\Delta_i, w_i, \mu_i, \gamma_i, \sigma_i^2\}$.

The sufficient conditions for $\hat{\theta}$ are obtained using the Lagrange multiplier method. We define the Lagrangian

$$\begin{aligned} \mathcal{L}_1 = & \sum_{j=1}^N y_j^{\text{nb}} \ln \sum_{i=1}^m w_i \left[\frac{\Delta_i}{\pi} \frac{\gamma_i}{(x_j - \mu_i)^2 + \gamma_i^2} + \frac{1 - \Delta_i}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{(x_j - \mu_i)^2}{2\sigma_i^2}\right) \right] \\ & + \lambda \left(\sum_{i=1}^m w_i - I_{\text{area}} \right), \end{aligned} \quad (26)$$

where λ is the Lagrange multiplier.

Taking first-order derivatives with respect to each parameter yields

$$\frac{\partial \mathcal{L}_1}{\partial w_i} = \sum_{j=1}^N y_j^{\text{nb}} \frac{\frac{\Delta_i}{\pi} \frac{\gamma_i}{(x_j - \mu_i)^2 + \gamma_i^2} + \frac{1 - \Delta_i}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{(x_j - \mu_i)^2}{2\sigma_i^2}\right)}{\sum_{i=1}^m w_i p_{\text{pv},k}(x_j)} + \lambda = 0, \quad (27)$$

$$\frac{\partial \mathcal{L}_1}{\partial \mu_i} = \sum_{j=1}^N y_j^{\text{nb}} \frac{w_i \left[\frac{\Delta_i}{\pi} \frac{2\gamma_i}{[(x_j - \mu_i)^2 + \gamma_i^2]^2} + \frac{1 - \Delta_i}{\sqrt{2\pi}\sigma_i^3} \exp\left(-\frac{(x_j - \mu_i)^2}{2\sigma_i^2}\right) \right]}{\sum_{i=1}^m w_i p_{\text{pv},k}(x_j)} (x_j - \mu_i) = 0, \quad (28)$$

$$\frac{\partial \mathcal{L}_1}{\partial \gamma_i} = \sum_{j=1}^N y_j^{\text{nb}} \frac{w_i \frac{\Delta_i}{\pi} \frac{(x_j - \mu_i)^2 - \gamma_i^2}{[(x_j - \mu_i)^2 + \gamma_i^2]^2}}{\sum_{i=1}^m w_i p_{\text{pv},k}(x_j)} = 0, \quad (29)$$

$$\frac{\partial \mathcal{L}_1}{\partial \sigma_i} = \sum_{j=1}^N y_j^{\text{nb}} \frac{w_i \frac{1 - \Delta_i}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{(x_j - \mu_i)^2}{2\sigma_i^2}\right)}{\sum_{i=1}^m w_i p_{\text{pv},k}(x_j)} \left(\frac{(x_j - \mu_i)^2}{\sigma_i^3} - \frac{1}{\sigma_i} \right) = 0, \quad (30)$$

$$\frac{\partial \mathcal{L}_1}{\partial \Delta_i} = \sum_{j=1}^N y_j^{\text{nb}} \frac{w_i \left[\frac{1}{\pi} \frac{\gamma_i}{(x_j - \mu_i)^2 + \gamma_i^2} - \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{(x_j - \mu_i)^2}{2\sigma_i^2}\right) \right]}{\sum_{i=1}^m w_i p_{\text{pv},k}(x_j)} = 0. \quad (31)$$

Solving Eq. (27) yields

$$w_i^{(t+1)} = I_{\text{area}} \frac{\sum_{j=1}^N y_j^{\text{nb}} T_{ji}^{(t)}}{\sum_{j=1}^N y_j^{\text{nb}}}, \quad (32)$$

which has a clear physical interpretation as the mixing probability of the i -th pseudo-Voigt component. Interestingly, the remaining parameters $\{\Delta_i, \mu_i, \gamma_i, \sigma_i^2\}$ can be updated directly in the M-step by incorporating additional information beyond $T_{ji}^{(t)}$ to obtain a “better” solution by our mathematical construction.

Specifically, we substitute the parameters $\theta^{(t)} = \{\Delta_i^{(t)}, \mu_i^{(t)}, \gamma_i^{(t)}, \sigma_i^{2(t)}\}$ into Eqs. (28)–(31), which significantly simplifies the optimization at the $(t+1)$ -th iteration. The resulting first-order conditions become

$$\frac{\partial \mathcal{L}_1}{\partial \mu_i} = \sum_{j=1}^N y_j^{\text{nb}} \left[\frac{2\pi}{\gamma_i} \beta_{ji}^L + \frac{1}{\sigma_i^2} T_{ji}^R \right] (x_j - \mu_i) = 0, \quad (33)$$

$$\frac{\partial \mathcal{L}_1}{\partial \gamma_i} = \frac{\pi}{\gamma_i^2} \sum_{j=1}^N y_j^{\text{nb}} \beta_{ji}^L (x_j - \mu_i)^2 - \pi \sum_{j=1}^N y_j^{\text{nb}} \beta_{ji}^L = 0, \quad (34)$$

$$\frac{\partial \mathcal{L}_1}{\partial \sigma_i} = \sum_{j=1}^N y_j^{\text{nb}} T_{ji}^R \left(\frac{(x_j - \mu_i)^2}{\sigma_i^3} - \frac{1}{\sigma_i} \right) = 0, \quad (35)$$

$$\frac{\partial \mathcal{L}_1}{\partial \Delta_i} = \sum_{j=1}^N y_j^{\text{nb}} \left(\frac{T_{ji}^L}{\Delta_i} - \frac{T_{ji}^R}{1 - \Delta_i} \right) = 0. \quad (36)$$

Consequently, the closed-form update rules in the $(t+1)$ -th maximization step are given by

$$\mu_i^{(t+1)} = \frac{\frac{2\pi}{\gamma_i^{(t)}} \sum_{j=1}^N y_j^{\text{nb}} \beta_{ji}^{L(t)} x_j + \frac{1}{\sigma_i^{2(t)}} \sum_{j=1}^N y_j^{\text{nb}} T_{ji}^{R(t)} x_j}{\frac{2\pi}{\gamma_i^{(t)}} \sum_{j=1}^N y_j^{\text{nb}} \beta_{ji}^{L(t)} + \frac{1}{\sigma_i^{2(t)}} \sum_{j=1}^N y_j^{\text{nb}} T_{ji}^{R(t)}}, \quad (37)$$

$$\gamma_i^{(t+1)} = \left[\frac{\sum_{j=1}^N y_j^{\text{nb}} \beta_{ji}^{L(t)} (x_j - \mu_i^{(t)})^2}{\sum_{j=1}^N y_j^{\text{nb}} \beta_{ji}^{L(t)}} \right]^{1/2}, \quad (38)$$

$$\sigma_i^{2(t+1)} = \frac{\sum_{j=1}^N y_j^{\text{nb}} T_{ji}^{R(t)} (x_j - \mu_i^{(t)})^2}{\sum_{j=1}^N y_j^{\text{nb}} T_{ji}^{R(t)}}, \quad (39)$$

$$\Delta_i^{(t+1)} = \frac{\sum_{j=1}^N y_j^{\text{nb}} T_{ji}^{L(t)}}{\sum_{j=1}^N y_j^{\text{nb}} T_{ji}^{(t)}}. \quad (40)$$

In each EM–Bragg iteration, the derived peak locations are correlated with a specific set of lattice constants by minimizing the following objective function:

$$\mathcal{L}_2 = \frac{1}{2m'} \sum_{i \in m'} \left(\mu_i^{\text{WPEM}} - 2 \arcsin \left(\frac{\lambda}{2d_{H_i K_i L_i}} \right) \right)^2, \quad (41)$$

where m' denotes a subset of stable diffraction peaks selected for lattice refinement, λ is the X-ray wavelength, and $d_{H_i K_i L_i}$ is the interplanar spacing associated with the $(H_i K_i L_i)$ reflection. After this correction, all peak positions are unified and strictly calibrated according to Bragg's law.

A simple gradient-descent scheme is employed to solve Eq. (41) for lattice calibration, yielding

$$\text{lc}^* = \text{lc} - r \frac{1}{m'} \sum_{i \in m'} \left(\mu_i^{\text{WPEM}} - \mu_i^{\text{Brg}} \right) \frac{\lambda}{d_{H_i K_i L_i}^2 \sqrt{1 - \lambda^2 / (4d_{H_i K_i L_i}^2)}} \frac{\partial d_{H_i K_i L_i}}{\partial \text{lc}}, \quad (42)$$

where $lc \in \{a, b, c, \alpha, \beta, \gamma\}$ denotes the lattice constants and lattice angles, and $\mu_i^{\text{Brg}} = 2 \arcsin(\lambda/2d_{H_i K_i L_i})$ is the Bragg peak position computed from the current lattice parameters. The learning rate r is estimated using a finite-difference scheme,

$$r = \frac{2\tau}{\frac{\partial \mathcal{L}_2(\mu_i^{\text{WPEM}}, \mu_i^{\text{Brg}}; lc + \tau)}{\partial (lc + \tau)} - \frac{\partial \mathcal{L}_2(\mu_i^{\text{WPEM}}, \mu_i^{\text{Brg}}; lc - \tau)}{\partial (lc - \tau)}}, \quad (43)$$

where the step size is set to $\tau = 0.05$. The optimization terminates when $|lc^* - lc| < 10^{-8}$. The updated lattice constants are subsequently used to compute Bragg peak positions for the next EM–Bragg iteration.

For multi-component (materials) mixtures, peaks from all components are first pooled and sorted by diffraction angle, while their component labels (peak identities) are stored for later retrieval. In the maximization step, all peaks are updated within the unified framework of Eqs. 37–40. The updated peak positions are then mapped back to each component and adjusted using its own lattice geometry via Eq. 42. Peaks belonging to the same crystalline component are grouped into batches, within which lattice parameters are refined independently by enforcing Bragg’s law. The optimized peak positions are finally mapped back to their original stored indices, propagated to the next EM–Bragg iteration, and used for subsequent assignment, as illustrated in Fig. 1. This mixed-phase polycrystalline powder-diffraction workflow is termed *batch EM–Bragg optimization*.

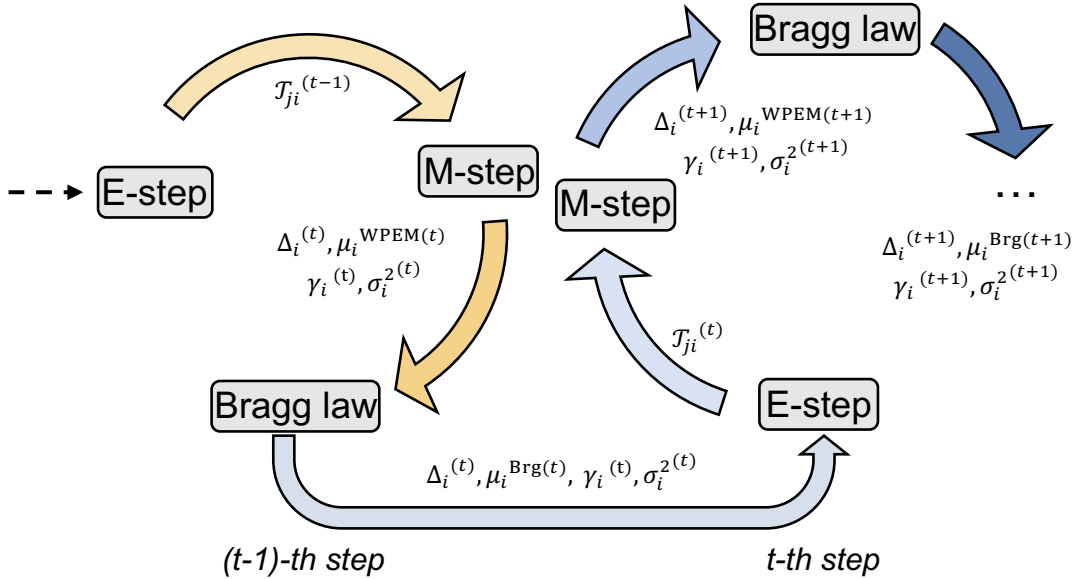


Figure 1. One EM–Bragg iteration. Given the background-subtracted diffraction profile and the current lattice constants, the E-step evaluates soft assignments (responsibilities) for each peak component, and the M-step updates peak-shape/position parameters while enforcing Bragg’s law through lattice-constant refinement.

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1.6 Volume Fraction Determination

The scattering of X-rays or neutrons by matter originates from their interaction with constituent atoms. For X-rays, scattering arises from the interaction between incident photons and the electrons

surrounding atomic nuclei². According to Thomson scattering theory, the intensity scattered by an ideal electron at a distance R is given by

$$I = I_0 \frac{e^4}{m^2 c^4 R^2} \sin^2 \varphi, \quad (44)$$

where I_0 is the incident X-ray intensity, e , m , and c denote the electron charge, mass, and the speed of light, respectively, and φ is the angle between the scattered X-ray direction and the electric field vector of the incident radiation.

For unpolarized incident X-rays, the electric field direction is arbitrary within the plane perpendicular to the incident beam. The corresponding scattered intensity of a single electron becomes

$$I = I_0 \frac{e^4}{m^2 c^4} \frac{1 + \cos^2 2\theta}{2}, \quad (45)$$

Here, 2θ denotes the scattering angle, and the factor $(1 + \cos^2 2\theta)/2$ represents the polarization term. **In the derivation of the EM–Bragg procedure, we introduce $\tilde{\theta}$ to distinguish the notation used for model parameters. For clarity and consistency with common diffraction notation, we subsequently use θ to denote the diffraction angle throughout the remainder of this Supporting Information.**

The atomic arrangement of a crystalline material can be described as an infinitely periodic repetition of a unit cell in three-dimensional space. The structure factor is therefore introduced to characterize the diffraction amplitude of a unit cell relative to a single electron and is expressed as

$$F_{HKL} = \sum_{j=1}^n f_j [\cos(2\pi(Hx_j + Ky_j + Lz_j)) + i \sin(2\pi(Hx_j + Ky_j + Lz_j))], \quad (46)$$

where f_j is the atomic scattering factor of the j -th atom, and (x_j, y_j, z_j) are its fractional coordinates within the unit cell.

The observed diffraction peak intensity is a combined result of electron scattering, systematic extinction, diffraction geometry, absorption, multiple scattering, and thermal vibration effects. It can be written as

$$I = \frac{1}{32\pi D} I_0 \frac{e^4}{m^2 c^4} \frac{\lambda^3}{V_0^2} V F_{HKL}^2 P \frac{1 + \cos^2 2\theta}{\sin^2 \theta \cos \theta} e^{-2M} A(\theta), \quad (47)$$

where D is the specimen–detector distance, λ is the X-ray wavelength, V_0 is the unit-cell volume, V is the irradiated powder volume, P is the reflection multiplicity, e^{-2M} is the Debye–Waller factor accounting for thermal vibration, and $A(\theta)$ denotes the absorption factor.

Based on diffraction intensity theory, the volume fractions of mixture components can be determined quantitatively. Eq. (47) can be decomposed into an angle-independent constant and an angle-dependent term. For powder components measured under identical experimental conditions, the Debye–Waller factor e^{-2M} can be approximated as constant. The absorption factor is further simplified as $A(\theta) \approx c/\mu$, where c is a constant and μ is the linear absorption coefficient, which depends only on the intrinsic material properties.

Under these approximations, the diffraction intensity of component i can be expressed as

$$I_i = C Q_i V_i, \quad (48)$$

where C is an angle-independent constant and

$$Q_i = F_{HKL,i}^2 P_i \frac{1 + \cos^2 2\theta}{\sin^2 \theta \cos \theta} \frac{1}{\mu_i} \quad (49)$$

is a component-dependent factor determined by the crystal structure, atomic positions, diffraction angle, and physical properties of component i .

The volume fraction ratio between components i and j is therefore given by

$$\frac{V_i}{V_j} = \frac{I_i/Q_i}{I_j/Q_j}. \quad (50)$$

Since the structure factor F_{HKL}^2 depends on detailed atomic coordinates and scattering factors that are not always readily available, we provide three alternative schemes for volume fraction estimation:

1. *Strict intensity-based method*: the volume fraction is computed directly using Eq. (50).
2. *Approximate intensity-based method*: only the angular factor $(1 + \cos^2 2\theta)/(\sin^2 \theta \cos \theta)$ is retained, yielding $Q_i \approx (1 + \cos^2 2\theta)/(\sin^2 \theta \cos \theta)$.
3. *Integral intensity method*: the volume fraction is estimated from the integrated diffraction intensities,

$$\frac{V_i}{V_j} = \frac{\int I_i d\theta}{\int I_j d\theta}, \quad (51)$$

where $\int I_i$ and $\int I_j$ are the integrated areas of the decomposed diffraction components.

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1.7 Background Stripping

Let $f(t)$ be a general periodic function with period T defined in reciprocal space. Its Fourier series expansion is given by

$$f(t) = \sum_{n=-\infty}^{\infty} C_n e^{2\pi i n t / T}, \quad (52)$$

where $e^{2\pi i n t / T}$ forms a complete set of basis functions in the reciprocal domain, representing mixed sub-signals such as Compton scattering and amorphous background contributions. The corresponding Fourier coefficients are

$$C_n = \frac{1}{T} \int_0^T f(t) e^{-2\pi i n t / T} dt. \quad (53)$$

The Fast Fourier Transform (FFT)³ efficiently evaluates the discrete Fourier transform for a finite set of sampled data. For N equidistant sampling points in the reciprocal domain,

$$\{f(0), \dots, f(i2\pi/N), \dots, f((N-1)2\pi/N)\}, \quad (54)$$

the sampled signal can be expressed as a system of equations

$$\left\{ \begin{array}{l} f(0) = \sum_{n=-\infty}^{\infty} C_n e^{2\pi i n \cdot 0/T}, \\ \vdots \\ f((N-1)2\pi/N) = \sum_{n=-\infty}^{\infty} C_n e^{2\pi i n (N-1)2\pi/(NT)}. \end{array} \right. \quad (55)$$

For a finite superposition of mixed signals, Eq. (55) can be written in matrix form as

$$\mathbf{F}_N = \mathbf{W}_N \mathbf{C}_N, \quad (56)$$

where $\mathbf{F}_N = (f(0), \dots, f((N-1)2\pi/N))^T$ is the sampled signal vector, $\mathbf{C}_N = (C_0, \dots, C_N)^T$ is the Fourier coefficient vector, and $\mathbf{W}_N$ is the Fourier transformation matrix. The decomposition of the observed diffraction pattern into mixed sub-signals is therefore obtained as

$$\mathbf{C}_N = \mathbf{W}_N^{-1} \mathbf{F}_N,$$

while the inverse Fourier transform reconstructs the signal in the original domain. The mixed scattering signals are smoothed and selectively suppressed in the frequency domain through filtering.

To enhance the signal-to-noise ratio, the experimental diffraction pattern is further smoothed to suppress high-frequency noise while preserving Bragg reflection features. The Savitzky–Golay (S–G) filter⁴ is employed as a sliding polynomial smoothing technique based on local least-squares fitting. The filtered intensity is given by

$$y_j^* = \frac{1}{N} \sum_{i=-m}^m c_i y_{j+i}, \quad (57)$$

where y_{j+i} denotes the original diffraction intensity, y_j^* is the filtered value, c_i are the polynomial coefficients obtained from least-squares fitting, and $2m+1$ defines the window length.

After S–G filtering, both the enhanced intensity y_j^* and the original experimental intensity y_j are available at identical diffraction angles. The statistical variance of background fluctuations is evaluated using the sample variance

$$s^2 = \frac{1}{n} \sum_{j=1}^n (y_j^* - y_j)^2, \quad (58)$$

where n is the total number of background sampling points.

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1.8 Amorphous Materials

Amorphous polymers, as well as amorphous regions in semicrystalline polymers, give rise to diffuse X-ray scattering⁵. When polymer chains exhibit preferred orientation, this diffuse halo may further split into equatorial arcs. Accurate separation of amorphous and crystalline scattering is therefore essential for probing disordered regions and for quantitatively determining crystallinity in semicrystalline polymers⁶.

For certain polymers, such as PET and PA-6, amorphous reference patterns can be obtained experimentally by rapidly quenching the melt to produce fully amorphous specimens. However, for

many polymers, it is difficult to prepare completely amorphous samples. In such cases, scattering from molten polymers cannot serve as a reliable amorphous template, as thermal expansion and changes in interchain distances significantly alter the shape and position of the amorphous halo. Even when available, experimentally obtained amorphous templates generally provide only a rough initial estimate of the amorphous scattering contribution⁵.

Here, we adopt a data-driven strategy in which the amorphous scattering is extracted directly from highly crystalline diffraction patterns by decomposing the total intensity into crystalline peaks and amorphous halos. The amorphous contribution, including its shape, position, and half-width, is modeled using a Gaussian mixture model, extending conventional amorphous fitting approaches⁷.

After subtracting the crystalline contribution, the amorphous halo intensity is obtained as

$$y_j^{\text{halo}} = y_j^{\text{ex}} - y_j^{\text{cry}}, \quad (59)$$

where y_j^{ex} denotes the experimental intensity and y_j^{cry} represents the crystalline component. Within the EM-Bragg optimization framework, the amorphous halo is treated as a new mixture measure function,

$$y_j^{\text{halo}} = \sum_{i=1}^k w_i p_i(x_j), \quad (60)$$

where $p_i(x_j)$ is a Gaussian probability density function,

$$p_i(x_j) = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left[-\frac{(x_j - \mu_i)^2}{2\sigma_i^2}\right], \quad (61)$$

w_i is the corresponding weight, and k is the number of amorphous components. The value of k is kept minimal, increasing from 1 up to a maximum of 4 only when the fitting quality is insufficient.

The total integrated intensity of the amorphous halo is defined as

$$\Omega(y^{\text{halo}}) = \int_{-\infty}^{+\infty} y^{\text{halo}}(x) dx = I_{\text{area}}^{\text{halo}}, \quad (62)$$

which imposes the constraint

$$\sum_{i=1}^k w_i = I_{\text{area}}^{\text{halo}}, \quad w_i \in (0, I_{\text{area}}^{\text{halo}}). \quad (63)$$

Accordingly, the $Q(\theta | \theta^{(t)})$ function for amorphous fitting is written as

$$\begin{aligned} Q(\theta | \theta^{(t)}) &= \ln \prod_{j=1}^N \left(\sum_{i=1}^k w_i p_i(x_j) \right)^{y_j^{\text{halo}}} \\ &= \sum_{j=1}^N y_j^{\text{halo}} \ln \left[\sum_{i=1}^k w_i \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{(x_j - \mu_i)^2}{2\sigma_i^2}\right) \right], \end{aligned} \quad (64)$$

where parameter $\theta = \{w_i, \mu_i, \sigma_i^2\}$.

The posterior responsibility is defined as

$$\xi_{ij}^{(t)} = \frac{w_i^{(t)} \frac{1}{\sqrt{2\pi\sigma_i^{2(t)}}} \exp\left[-\frac{(x_j - \mu_i^{(t)})^2}{2\sigma_i^{2(t)}}\right]}{\sum_{i=1}^k w_i^{(t)} \frac{1}{\sqrt{2\pi\sigma_i^{2(t)}}} \exp\left[-\frac{(x_j - \mu_i^{(t)})^2}{2\sigma_i^{2(t)}}\right]}.$$

Maximization with respect to parameter θ under the constraint $\sum_{i=1}^k w_i = I_{\text{area}}^{\text{halo}}$ is performed using the Lagrange multiplier method, yielding

$$\mathcal{L}_3 = \sum_{j=1}^N y_j^{\text{halo}} \ln \left(\sum_{i=1}^k w_i p_i(x_j) \right) + \rho \left(\sum_{i=1}^k w_i - I_{\text{area}}^{\text{halo}} \right). \quad (65)$$

The parameter updates at iteration $(t + 1)$ are given by

$$w_i^{(t+1)} = I_{\text{area}}^{\text{halo}} \frac{\sum_{j=1}^N y_j^{\text{halo}} \xi_{ij}^{(t)}}{\sum_{j=1}^N y_j^{\text{halo}}}, \quad (66)$$

$$\mu_i^{(t+1)} = \frac{\sum_{j=1}^N y_j^{\text{halo}} \xi_{ij}^{(t)} x_j}{\sum_{j=1}^N y_j^{\text{halo}} \xi_{ij}^{(t)}}, \quad (67)$$

$$\sigma_i^{2(t+1)} = \frac{\sum_{j=1}^N y_j^{\text{halo}} \xi_{ij}^{(t)} (x_j - \mu_i)^2}{\sum_{j=1}^N y_j^{\text{halo}} \xi_{ij}^{(t)}}. \quad (68)$$

Following this iterative Gaussian mixture optimization, the amorphous halo can be quantitatively characterized in a self-consistent manner.

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1.9 Evaluation of Refinement Quality

The quality of profile fitting is evaluated using the weighted profile factor (R_{wp}) and the profile factor (R_p)⁸:

$$R_{\text{wp}} = 100 \sqrt{\frac{\sum_{j=1}^N \hat{w}_j (\hat{y}_j - y_j^{\text{ex}})^2}{\sum_{j=1}^N \hat{w}_j (y_j^{\text{ex}})^2}}, \quad R_p = \frac{\sum_{j=1}^N |\hat{y}_j - y_j^{\text{ex}}|}{\sum_{j=1}^N y_j^{\text{ex}}}. \quad (69)$$

Here, y_j^{ex} and $\hat{y}_j$ denote the experimentally observed and fitted intensities at $x = x_j$, respectively, and $\hat{w}_j = 1/y_j^{\text{ex}}$ is the statistical weight of the j -th data point. To ensure that all values of the background-subtracted profile y_j^{nb} remain positive, the refinement is performed after adding a constant baseline defined by the minimum intensity of the no-background profile, $y_{\text{min}}^{\text{nb}}$. Accordingly, the fitted profile is expressed as

$$\hat{y}_j = y_j^{\text{bac}} + y_j^{\text{WPEM}} + y_{\text{min}}^{\text{nb}}. \quad (70)$$

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1.10 Convergence

Several truncation conditions are incorporated into our strategy to ensure numerical stability. The convergence criterion in the EM–Bragg process is defined by the difference between the log-likelihood values of two consecutive iterations, which is required to be smaller than a prescribed threshold δ (default $\delta = 0.05$). If this condition is not satisfied, the updated values of Δ_i or $1 - \Delta_i$ may approach the limits of machine precision, potentially leading to numerical instability and delayed convergence. To prevent this, the iteration is suspended whenever $\Delta_i \leq \epsilon$ or $(1 - \Delta_i) \leq \epsilon$, where ϵ is set to 10^{-9} by default.

Excessive iterations are physically meaningless for lattice-constant determination because they imply unrealistically high numerical precision. Accordingly, the our code is terminated once the number of iterations exceeds a predefined maximum (default 300). During each iteration, the evolution of the log-likelihood, the weighted profile factor R_{wp} , and the profile factor R_p is recorded in a trace file to facilitate convergence diagnostics and ensure result reliability. In practice, a nearly stable solution is typically achieved within approximately 10–140 iterations.

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1.11 Python Package: PyXplore

The underlying algorithmic framework is nontrivial, involving iterative EM–Bragg optimization, peak mixture modeling, and constrained parameter estimation. To facilitate reproducible application and broader adoption, we developed a lightweight and user-friendly software toolkit, PyXplore, that implements the complete workflow described above.

PyXplore is an open-source Python package organized into several core modules, each corresponding to a key stage of the whole powder pattern decomposition and EM–Bragg optimization pipeline. These modules provide functionality for diffraction data preprocessing, background estimation and subtraction, crystalline peak detection and modeling, amorphous halo decomposition, constrained Gaussian mixture optimization, convergence control, and quantitative post-analysis, including peak integration and crystallinity evaluation.

The source code is publicly available at <https://github.com/Bin-Cao/PyWPEM>, and comprehensive documentation, usage examples, and a detailed methodological description are provided at <https://pyxlore.netlify.app>.

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2 Comparison with FullProf and TOPAS

Figs. 2 and 3 present the Le Bail fitting results for $\text{Tb}_2\text{BaCoO}_5$ and PbSO_4 obtained using the *profile matching with constant scale factor* approach implemented in the FullProf package.

In the TOPAS package, full-pattern fitting is performed using the fundamental parameters approach (FPA), in which the diffraction profile is modeled through direct convolution rather than a pseudo-Voigt function. The resulting intensity profile is expressed as

$$y(x) = [W(x) \otimes G(x)] \otimes S(x), \quad (71)$$

where $\otimes$ denotes convolution, and $W(x)$, $G(x)$, and $S(x)$ represent the emission profile, instrumental contributions, and specimen-related effects, respectively. The corresponding PSF is shown in Fig.

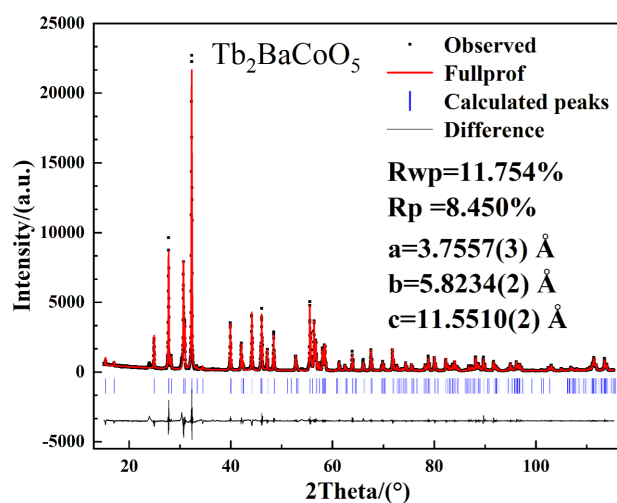


Figure 2. Profile fitting using FullProf for $\text{Tb}_2\text{BaCoO}_5$. Black dots: observed profile. Red curve: calculated fit.

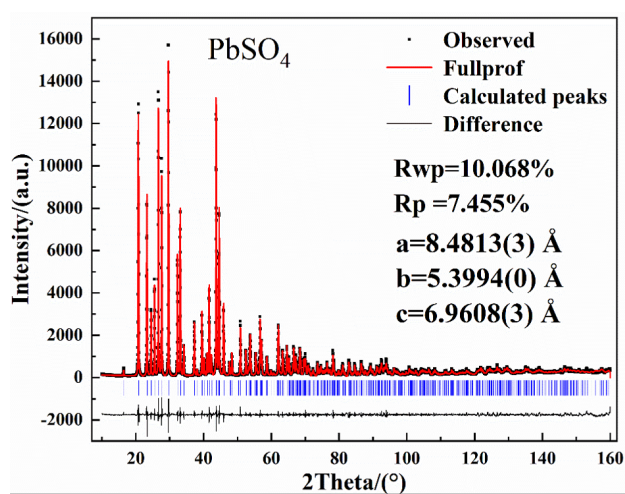


Figure 3. Profile fitting using FullProf for PbSO_4 . Black dots: observed profile. Red curve: calculated fit.

4. Fig. 5 displays the Le Bail fitting result for PbSO_4 obtained using this convolution-based PSF in TOPAS.

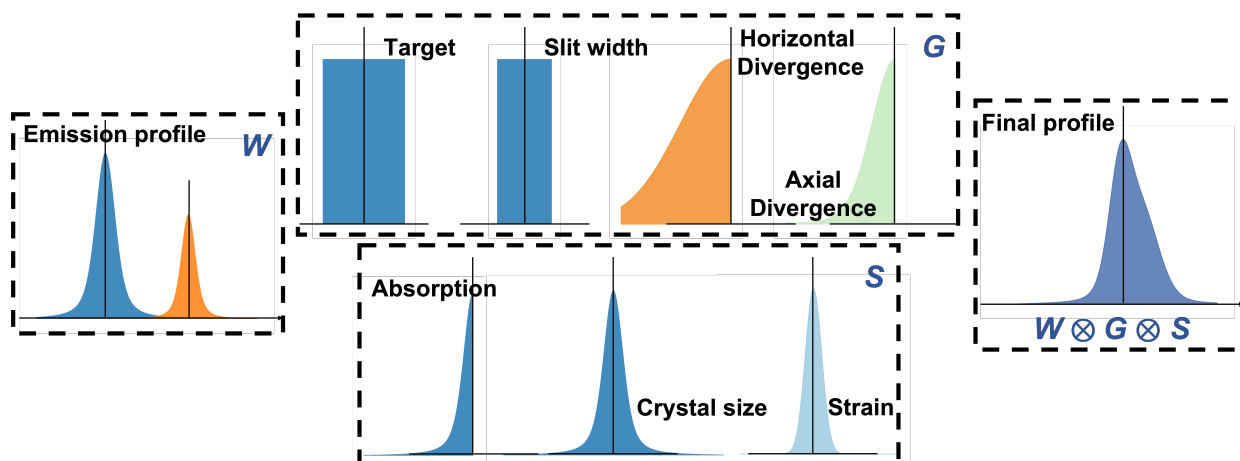


Figure 4. Fundamental-parameters approach (FPA) peak-shape function. Emission profile (W), instrumental factor (G) and specimen contribution (S) are convolved to form the PSF.

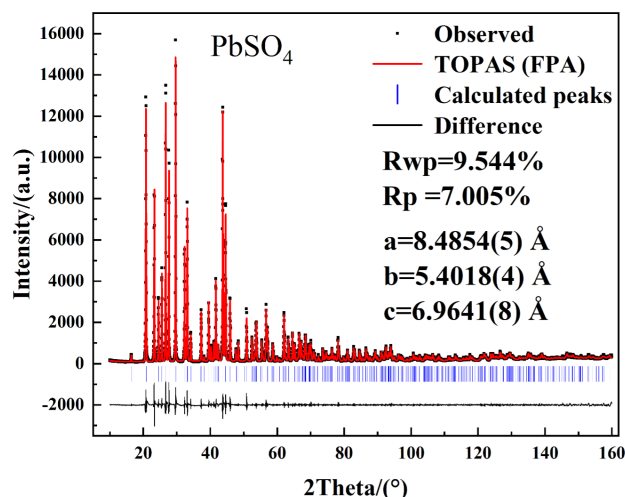


Figure 5. Profile fitting using TOPAS for PbSO_4 . Black dots: observed profile. Red curve: calculated fit.

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3 Crystal Systems and Inter-planar Spacing

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4 Structure determination workflow

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4.1 In-library and out-of-library tasks

In-library identification is a foundational task in powder crystallography, aiming to recognize phases by matching an experimental XRD/PXRD pattern against curated reference collections.

Table 1. Lattice parameters of the seven crystal systems.

Code	Crystal system	Lattice parameters
1	Cubic	$a = b = c, \alpha = \beta = \gamma = 90^\circ$
2	Hexagonal	$a = b \neq c, \alpha = \beta = 90^\circ, \gamma = 120^\circ$
3	Tetragonal	$a = b \neq c, \alpha = \beta = \gamma = 90^\circ$
4	Orthorhombic	$a \neq b \neq c, \alpha = \beta = \gamma = 90^\circ$
5	Trigonal	$a = b = c, \alpha = \beta = \gamma \neq 90^\circ$
6	Monoclinic	$a \neq b \neq c, \alpha = \gamma = 90^\circ, \beta \neq 90^\circ$
7	Triclinic	$a \neq b \neq c, \alpha \neq \beta \neq \gamma, \alpha, \beta, \gamma \neq 90^\circ$

Table 2. Interplanar spacing d_{hkl} for the seven crystal systems.

Crystal system	Interplanar spacing d_{hkl}
Cubic	$\frac{a}{\sqrt{H^2 + K^2 + L^2}}$
Hexagonal	$\frac{\sqrt{3}/2}{\sqrt{(H^2 + K^2 + HK)/a^2 + 3L^2/(4c^2)}}$
Tetragonal	$\frac{1}{\sqrt{(H^2 + K^2)/a^2 + L^2/c^2}}$
Orthorhombic	$\frac{1}{\sqrt{H^2/a^2 + K^2/b^2 + L^2/c^2}}$
Trigonal	$a \sqrt{\frac{1 - 3 \cos^2 \alpha + 2 \cos^3 \alpha}{(H^2 + K^2 + L^2) \sin^2 \alpha + 2(HK + KL + HL) \cos^2 \alpha - \cos \alpha}}$
Monoclinic	$\frac{ \sin \beta }{\sqrt{H^2/a^2 + K^2/b^2 \sin^2 \beta + L^2/c^2 - 2HL/(ac) \cos \beta}}$
Triclinic	$d = \frac{m}{\sqrt{(H/a)^2 \sin^2 \alpha + (K/b)^2 \sin^2 \beta + (L/c)^2 \sin^2 \gamma - 2HK/(ab)q - 2HL/(ac)n - 2KL/(bc)p}}$

where : $m = \sqrt{\sin^2 \gamma - \cos^2 \alpha + \cos^2 \beta + 2 \cos \alpha \cos \beta \cos \gamma}$, $p = \cos \alpha - \cos \beta \cos \gamma$,
 $n = \cos \beta - \cos \alpha \cos \gamma$, and $q = \cos \gamma - \cos \alpha \cos \beta$.

Since the establishment of standardized powder diffraction databases (e.g., the Powder Diffraction File, PDF) in 1938, crystallographers have continuously documented known phases together with their characteristic peak positions and relative intensities. In practice, an experimental pattern is pre-processed (background subtraction, smoothing, peak finding), converted into a compact set of peak descriptors, and then queried against the database to retrieve candidate phases. With modern computing, this workflow has been implemented in a wide range of search–match programs that differ mainly in peak extraction, scoring functions, and how mixtures are handled.

By contrast, out-of-library identification targets phases whose structures are absent from databases, and therefore cannot be solved by retrieval. In this setting, the goal is to infer a consistent crystallographic model directly from data: determining the lattice metric and symmetry, proposing plausible atomic configurations that satisfy chemistry and symmetry, and validating them by refinement against the measured pattern. A common route is: (i) indexing to obtain lattice parameters from peak positions; (ii) space-group determination using systematic absences and intensity statistics; (iii) composition constraints from chemistry, complementary measurements (e.g., elemental analysis), or plausible stoichiometries; and (iv) structure solution (direct methods, simulated annealing, global optimization) followed by Rietveld refinement. Throughout this process, the reliability of peak positions and integrated intensities is critical: severe overlap, preferred orientation, microstrain, and multiphase mixtures can easily destabilize the inversion from a one-dimensional profile to a three-dimensional structure.

More recently, AI has started to play a role in both in-library and out-of-library settings. For in-library tasks, the dominant paradigm is *classification*: a model maps an input pattern (or derived features) to one or more labels corresponding to candidate phases, thereby *unlocking* conventional retrieval and refinement pipelines with fast, robust candidate screening. For out-of-library tasks, AI is more naturally framed as *generation*: models propose physically plausible downstream structural hypotheses (e.g., symmetry/lattice candidates or atomic configurations) that can be validated and refined against diffraction data, helping bridge the gap between a one-dimensional profile and a three-dimensional structure when no database entry exists.

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4.2 Search-Match approach

The predominant strategy for in-library phase identification is search–match, which converts a measured pattern into a set of peak descriptors and then retrieves the closest entries from a reference database. A typical workflow involves three steps. First, a peak list is extracted from the profile and represented as (d, I) pairs, where d denotes the interplanar spacing and I the peak intensity (or an integrated area). Second, candidate phases are retrieved by searching for matching d values, historically aided by the Hanawalt indexing scheme. Third, candidates are ranked by a scoring function that accounts for both matched and unmatched peaks, as well as intensity consistency, to identify the best explanation of the data. The procedure may be repeated iteratively to account for mixtures, by subtracting a matched phase and re-running the search on the residual.

In-library retrieval provides a strong “warm start” for subsequent refinement: database entries supply lattice geometry and often approximate atomic coordinates, which can be refined against the experimental data. However, search–match is inherently limited by database coverage and can be ambiguous in the presence of peak overlap, similar phases, or mixtures. For out-of-library structures, one must instead infer symmetry and lattice parameters from the pattern itself and then

perform structure solution and refinement, which makes the accurate treatment of overlapped peaks and the enforcement of diffraction consistency particularly important.

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4.3 Whole powder pattern fitting and refinement

Refinement plays a crucial role in establishing crystal structures from powder data. Starting from an initial model (lattice parameters, atomic positions, profile/peak-shape and instrument parameters), refinement adjusts these parameters to minimize the discrepancy between calculated and measured intensities. Without a reasonable starting model, refinement can become ill-conditioned and may converge to non-unique or non-physical solutions, especially when reflections overlap strongly.

Identifying the crystal type begins the process, but its validation relies on the refinement results. The Rietveld method marked a significant advancement in powder diffraction fitting for structure refinement by introducing whole-pattern fitting, replacing the analysis of individual, non-overlapping Bragg peaks. The approach employs profile intensity calculations and a least-squares algorithm for refinement. By minimizing discrepancies between experimental and theoretical profiles, it extracts all structural information. Once the fitting process converges, the parameters in the theoretical model represent the only determinable physical quantities. Subsequently, the Pawley and Le Bail methods, two widely used whole powder pattern fitting techniques, are developed following the Rietveld method.

Practical powder refinement must also account for non-ideal experimental effects that can bias intensities and peak positions, including preferred orientation, anisotropic size/strain broadening, absorption, and sample displacement/zero shift. These effects are typically treated by adding physically motivated correction terms or additional profile parameters, which increases model flexibility but also raises the risk of overfitting when the data are noisy or heavily overlapped. This motivates the development of refinement workflows that enforce hard diffraction constraints (e.g., Bragg-consistent peak centres) while providing statistically stable intensity partitioning across overlapping reflections—the central goal of the WPEM framework described in the main text.

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